MIL-P-50202A (AR)
11 April 1977
SUPERSEDING
MIL-P-50202 (PA)
23 September 1974

MILITARY SPECIFICATION

PROPELLANT CHARGE, ROCKET, FOR DEMOLITION KIT, CRATERING, M180

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not normally covered by the drawings and provides quality assurance provisions for the fabrication of parts, the manufacture of the propellant and the assembly of one type of propellant charge for the Rocket Motor for Demolition Kit Cratering, M180.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents.-The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-14737 - Propellant M-7
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-453 - Radiographic Inspection

FSC: 1340

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (see 6.5)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9244409 - Propellant, Charge, Rocket

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transporation, parts 100-199.

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders shall specify 49 CFR 100-199.

3. REQUIREMENTS

3.1 Materials.-Materials shall be in accordance with the applicable drawings and specifications.

3.1.1 Propellant.-The propellant shall comply with specification MIL-P-14737, Type II.

3.2 Components and assemblies.-The components and assemblies shall comply with all requirements specified on drawing 9244409 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Radiographic quality.-X-rays for gap or separation evaluation shall be taken in accordance with MIL-STD-453 and shall be of a quality which will permit proper evaluation of the sealing quality as determined by the responsible technical agency of the Government. Minimum optical density of each radiograph in the area being inspected shall not be less than 1.25 and shall be within a density range that will permit proper identification of gap or separation defects.

3.4 Strength of propellant charge, rocket.-The Propellant grain shall not separate from the Molded Stud Assembly when subjected to a minimum axial force of 150 lbs.

3.5 Static functioning test

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3.5.1 Maximum pressure at 130°F.-The maximum pressure developed by the Propellant Charge Assembly at 130°F shall not exceed 12,000 pounds per square inch (p.s.i.) when fired in the applicable test equipment.

3.5.2 Action time at minus 40°F.-The action time developed by the Propellant Charge Assembly at minus 40°F shall not exceed 0.052 second.

3.5.3 Total impulse at minus 40°F.-The total impulse developed by the Propellant Charge Assembly at minus 40°F shall not be less than 900 pound seconds.

3.5.4 Ignition delay time at minus 40°F.-The ignition delay time at minus 40°F shall not exceed 0.025 second.

3.6 Propellant charge, rocket.-Any gap or separation at the Molded Stud Assembly and Grain Propellant, Machined interface shall be completely sealed with specific inhibitor so as to prevent burning of the stick at this interface.

3.7 First article inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Workmanship

3.8.1 Workmanship-Propellant grain.-The propellant grain shall be free of dirt, grease, chips, gouges, inhibitor solvent drippings, cracks, fissures, porous material and other defects which would make the item unsuitable for the intended use.

3.8.2 Workmanship-Parts and assemblies.-The parts and assemblies shall be free from chips, dirt, grease and other foreign matter that may impair proper functioning. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspections.-The following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part description	Drawings	Quantity
Propellant Charge, Rocket	9244409	15
Stud, Propellant, Rocket	9244420	10
Molded Stud Assembly	9244421	5 Consecutively produced from each cavity of each mold
Grain, Propellant, Machined	9270900	10
Propellant Charge Assembly	9244401 (Ref.)	*10

*Note: (See Dwg. 9244401) 72 Propellant charges., Dwg. 9244409 are required for one (1) Propellant Charge Assembly.

4.3.2 Inspections to be performed.-See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection.-See MIL-A-48078.

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TABLE 1 - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Propellant Charge Assembly				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Grain, Propellant, Machined (Dwg. 9270900)				
	Examination for Defects	10	-	3.2	4.4.2.1
	Stud Propellant, Rocket (Dwg. 9244420)				
	Examination for Defects	10	-	3.2	4.4.2.2
	Molded Stud Assembly (Dwg. 9244421)				
	Examination for Defects	5/ cavity	-	3.2	4.4.2.3
	Propellant Charge Rocket (Dwg. 9244409)				
	Examination for Defects	15	-	3.2	4.4.2.4
	Radiographic Examination for Assembly Gap or Separation Defects	15	-	3.6	4.5.3
	Radiographic Quality	15	-	3.3	4.5.3.1
	Strength of Propellant Charge, Rocket	15	-	3.4	4.5.4

NOTES:

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TABLE 1 - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	CONTINUATION SHEET	2 OF 2	See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Propellant Charge Assembly (See Dwg. 9244401) 72 Propellant Charges, Dwg. 9244409 are required for one (1) Propellant Charge Assembly.				
	Static Functioning Test at plus 130°F Maximum Pressure	5	-	3.5.1	4.5.5.1
	Static Functioning Test at minus 40°F Action Time	5	-	3.5.2	4.5.5.2
	Total Impulse	5	-	3.5.3	4.5.5.2
	Ignition Delay Time	5	-	3.5.4	4.5.5.2

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of Propellant Charge Assemblies shall contain:

a. Propellant grains from not more than one lot number from one manufacturer.

b. Inert parts from not more than one interfix lot number from one manufacturer.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Grain, Propellant, Machined		9270900
			NEXT HIGHER ASSEMBLY 9244409

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of counterbore		0.40%	3.2	Gage
102	Depth of counterbore		0.40%	3.2	Gage
103	Total length, Maximum (max.)		0.40%	3.2	Gage
104	Outside diameter (O.D) of grain, max.		0.40%	3.2	Gage
105	Grain cracked or damaged		0.40%	3.2	Visual
Minor					
201	Runout of counterbore with O.D.		0.65%	3.2	Gage
202	Roundness of O.D.		0.65%	3.2	Gage
203	Evidence of poor workmanship		1.00%	3.8.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Stud, Propellant Rocket	1 OF 1	9244420
			NEXT HIGHER ASSEMBLY 9244421

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of head		0.40%	3.2	Gage
102	Diameter of surface adjacent to head		0.40%	3.2	Gage
103	Large diameter of serrations		0.40%	3.2	Gage
104	True position of diameter of serrations with datum A		0.40%	3.2	Gage
105	Profile of serrations and angle of head taper		0.40%	3.2	Gage
106	Length from serrations to end		0.40%	3.2	Gage
107	Length of head land		0.40%	3.2	Gage
Minor					
201	Protective finish missing or damaged		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.8.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Molded Stud Assembly	1 OF 1	9244421
			NEXT HIGHER ASSEMBLY: 9244409

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length from end surface to shoulder		0.40%	3.2	Gage
102	Small diameter		0.40%	3.2	Gage
103	Mold inspection	5/ cavity		3.2	4.5.2
Minor					
201	True position of small diameter with datum A		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.00%	3.8.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50202A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Propellant Charge, Rocket	1 OF 1	9244409
			NEXT HIGHER ASSEMBLY 9244401

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Gap at stud assembly shoulder with propellant grain interface not completely sealed		0.40%	3.2	Visual
102	Radiographic examination for assembly gap or separation defects	(a)	0.40%	3.6	4.5.3
103	Strength of propellant charge, rocket	(b)	0.40%	3.4	4.5.4
Minor					
201	Evidence of poor workmanship		1.00%	3.8.2	Visual

NOTES:
(a) Sample size per MIL-STD-105, Inspection Level I.
(b) Sample size per MIL-STD-105, Special Inspection Level S-4.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Propellant Charge Assembly (72) Propellant Charges, Dwg. 9244409 are required for one (1) Propellant Charge Assembly	1 OF 1	9244401 REF.
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Static Functioning Test at 130°F Maximum pressure	(b)(c)	0.65%	3.5.1	4.5.5.1
	Static Functioning Test at minus 40°F				
102	Action Time	(c)	-	3.5.2	4.5.5.2
103	Total Impulse	(c)	-	3.5.3	4.5.5.2
104	Ignition Delay Time	(c)	-	3.5.4	4.5.5.2
Minor	None defined				

NOTES: (b) Sample size per MIL-STD-105, Special Inspection Level S-4.
(c) One half of the samples from (b) shall be used for functioning test at plus 130°F and the other half for functioning test at minus 40°F. One failure at either -40°F or +130°F rejects the lot.

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4.4.3 Testing

4.4.3.1 Testing.-Testing and examination of the samples for First Article and Quality Conformance are completely described in the First Article and Quality Conformance tables.

4.4.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures

4.5.1 Materials, components and processes.-Compliance with all requirements of section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections.-The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Molded stud assembly.-After production of 2000 Molded Stud Assemblies from each cavity of each mold, 5 stud assemblies per cavity per mold shall be consecutively produced and inspected for all drawing requirements. If any part fails to meet all drawing requirements, production shall be stopped and the mold corrected. During production, the molding procedures shall be monitored for uniformity of molding practices. Suspected deviations in mold product shall be examined to determine if corrective action or mold repair is required.

4.5.3 Radiograph examination for assembly gap or separation defects.-The samples of Rocket Propellant Charge selected shall be examined in accordance with MIL-STD-453 procedure. The items shall be X-rayed with the radiographic beam perpendicular to the longitudinal axis of the Machined Propellant Grain. The radiograph shall be examined to determine if any Molded Stud Assembly is not fully seated on the Machined Propellant Grain shoulder. The penetrameter used in the application specified in MIL-STD-453

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shall be designed and constructed to insure a 2 percent sensitivity or better. If any Rocket Propellant Charge shows evidence of the Molded Stud Assembly not fully seated on the Machined Propellant Grain counterbore interface as per requirement, it shall be classed defective.

4.5.3.1 Radiographic evaluation.-The Rocket Propellant Charge submitted as the First Article Samples, shall be radiographed in accordance with 4.5.3 and the radiographs shall be submitted to the Government for evaluation.

4.5.4 Strength of rocket propellant charge.-The samples selected for this test shall be conditioned at 70 degrees F plus or minus 5 degrees F for (6) hours minimum. The Rocket Propellant Charge shall be removed from the conditioning chamber and pulled to destruction within two (2) minutes after removal from the chamber. The test equipment shall be operated to exert a force at the rate of 2.5 inches per minute. If a failure occurs prior to attaining the specified strength requirements, the assembly shall be classed defective.

4.5.5 Static functioning test.-Propellant Charge Assembly, shall meet requirements for Maximum Pressure, Action Time, Total Impulse and Ignition Delay Time.

4.5.5.1 One half of the samples selected for this test shall be assembled to the applicable test fixture and equipment and conditioned at plus 130 degrees F plus or minus 3 degrees F for a minimum of six (6) hours. The propellant charges shall be fired within five minutes after removal from the conditioning chamber and the Maximum Pressure shall be obtained. If any propellant charge fails to meet the requirements, it shall be classed defective.

4.5.5.2 The other half of sample propellant charges shall be assembled to the applicable test fixture and equipment and conditioned at minus 40 degrees F plus or minus 3 degrees F for a minimum of six (6) hours. The propellant charges shall be fired within five minutes after removal from the conditioning chamber and the Action Time, Total Impulse and Ignition Delay Time shall be obtained. If any propellant charge fails to meet the requirements, it shall be classed defective.

5. Packaging

5.1 Preservation and packaging

5.1.1 Level C.-The propellant, charge, rocket shall be packaged in accordance with Code of Federal Regulations 49CFR parts 171-179.

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5.2 Packing

5.2.1 Level C.-The propellant, charge, rocket shall be packed in accordance with Code of Regulations 49CFR parts 171-179 and in a manner that will assure safe delivery from the manufacturers plant to the first receiving activity.

5.3 Marking.-The unit packages shall be marked in accordance with DOT Regulations and MIL-STD-129.

6. NOTES

6.1 Intended use.-The components covered by this specification are intended for use on the Demolition Kit, Cratering M180.

6.2 Ordering data.-See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval.-See MIL-A-48078. Submit equipment designs, as required, to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

6.4 Submission of test data.-When required by incorporation of DD Form 1423 into the contract, all test data required in Section 4 of this specification shall be submitted in accordance with Data Item Description DI-R-1724. In addition to the normal distribution of records, one copy of all test data shall be forwarded to each of the following:

Commander
US Army Armament Research and Development Command
ATTN: DRDAR-QAR-Q
and DRDAR-
Dover, New Jersey 07801

6.5 Drawings.-Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal Drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.6 Definitions

6.6.1 Ballistic definition.-Ballistic nomenclature shall be in accordance with MIL-STD-292.

6.7 Changes from previous issue.-Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Preparing Activity:
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Project Number: 1340-A528

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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